

$$\phi = \Phi(x) = \frac{1}{1+e^{-x}} = (1+e^{-x})^{-1}$$

$$\Rightarrow \frac{d\phi}{dx} = \frac{d(1+e^{-x})^{-1}}{dx}$$

$$= -(1+e^{-x})^{-2} \frac{d}{dx}(1+e^{-x})$$

$$= -(1+e^{-x})^{-2} \{0 + e^{-x} \cdot \frac{d}{dx}(-x)\}$$

$$= -(1+e^{-x})^{-2} \{e^{-x}(-1)\}$$

$$= e^{-x} (1+e^{-x})^{-2}$$

$$= \frac{e^{-x}}{1+e^{-x}} \times \frac{1}{1+e^{-x}}$$

$$= \frac{e^{-x} + 1 - 1}{1+e^{-x}} \times \phi \quad \because \phi = \frac{1}{1+e^{-x}}$$

$$= \left(1 - \frac{1}{1+e^{-x}}\right) \times \phi$$

$$= (1 - \phi) \phi$$